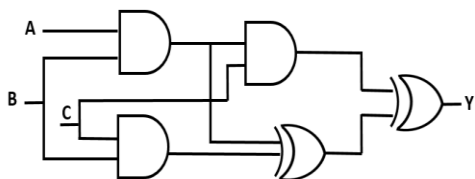




1. In order to store floating numbers in computers using the normalized representation and 32- bit single precision, the number of bits used for exponent and and fraction are\_\_\_\_\_,\_\_\_\_\_ respectively.

(a)11,21 (b)16,15  
(c)16,16 (d)8,23

2. The output of the combinational circuit given below is



(a)  $A + B + C$  (b)  $A(B + C)$   
(c)  $B(C + A)$  (d)  $C(A + B)$

3. What type of error are not detected by assemblers?

(a) Syntax error (b) Run-time error  
(c) Logical error (d) All of these

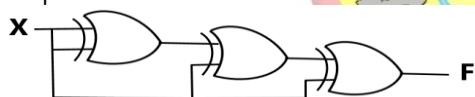
4. In IEEE single precision floating point representation, exponent is represented in.....

(a) 8 bit sign-magnitude representation  
(b) 8 bit 2's complement representation  
(c) Biases exponent representation with a bias value 127  
(d) Biases exponent representation with a bias value 128

5. If we can generate a maximum of 4 Boolean functions using n Boolean variables, what will be minimum value of n?

(a) 65536 (b) 16  
(c) 1 (d) 4

6. For the circuit shown below, the complement of the output



(a) 0 (b) X (c)  $X'$  (d) 1

7. How many Boolean expressions can be formed with 3 Boolean variables?

(a) 16 (b) 1024  
(c) 32 (d) 256

8. In an 8 bit representation of computer system the decimal number 47 has to be subtracted from 38 and the result in binary 2's complement is ...

(a) 11110111 (b) 10001001  
(c) 11111001 (d) 11110001

9. The memory unit which directly communicates with the CPU is known as

(a) Primary Memory (b) Secondary Memory  
(c) Shared Memory (d) Auxiliary Memory

10. Dynamic RAM consumes .....power and .....than Static RAM

(a) More, Faster (b) More, slower  
(c) Less, Slower (d) Less, Faster

11. The time required for fetching and execution of one simple machine instruction is known as

(a) Delay time (b) CPU cycle  
(c) Real time (d) Seek Time

12. The equivalence of given expression  $x + x'y$  with Boolean theorem is.....

(a) x (b)  $x + y$  (c)  $x'$  (d) 0

13. The logic XOR operation of  $(4AC0)_{16}$  and  $(B53F)_{16}$  results

(a) AACB (b) 0000  
(c) FFFF (d) ABCD

14. The process when processor fetch or decode another instruction during the execution of current instruction is called

(a) Supercomputing (b) Pipelining  
(c) Cloud computing (d) Grid computing

15. Which of the following is used by ALU to store the intermediate results?

(a) Stack (b) Heap  
(c) Registers (d) Accumulators

16. One TeraByte (TB)=..... GB and

One TeraByte (EB)=.....GB

(a)  $2^{10}GB, 2^{16}GB$  (b)  $2^{10}GB, 2^{20}GB$   
(c)  $2^{10}GB, 2^{24}GB$  (d)  $2^{10}GB, 2^{30}GB$

17. The cache memory is more effective because of

(a) Memory localization  
(b) Locality of reference  
(c) Memory size  
(d) None of the mentioned

18. Which of the following is the fastest means of memory access for CPU?

(a) Registers (b) cache  
(c) Main memory (d) Stack

19. To fetch data from the secondary memory which one of the following register is used?

(a) MAR (b) PC  
(c) IR (d) MBR

20. Consider a computer system with speed of  $10^6$  instructions per second. A program P, having  $2n^2$  steps is run on this system, where n is the input size. If  $n = 10000$ , what is the execution time for P?

(a) 1.2 (b) 20s  
(c) 100s (d) 200s

## ANSWER

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1  | d | 2  | c | 3  | c | 4  | c | 5  | c |
| 6  | d | 7  | d | 8  | a | 9  | a | 10 | c |
| 11 | b | 12 | b | 13 | c | 14 | b | 15 | d |
| 16 | d | 17 | b | 18 | a | 19 | a | 20 | d |





## Solution:

1. (d)

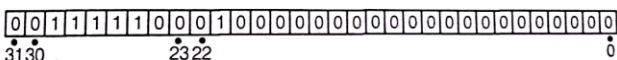
The IEEE 754 standard specifies a binary 32 bit as having:

Sign bit : 1 bit

Exponent width : 8 bits

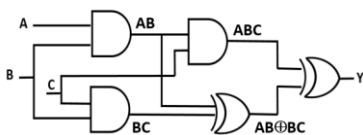
Significant precision : 24 bits (23 explicitly stored)

The bits are laid out as follows:



The real value assumed by a given 32-bit binary 32 data with a given biased sign, exponent  $e$  (the 8-bits unsigned integer) and a 23-bit fraction.

2. (c)



So,  $Y = ABC \oplus (AB \oplus BC)$

$Y = B(C + A)$

3. (c)

Logical error.

4. (c)

In IEEE single precision floating point representation, exponent is represented in Biases exponent representation. The biased exponent is used for representation of negative exponents.

The biased exponent ( $E$ ) of single precision number can be obtained as

$E = e + 127$

5. (c)

On  $n$  Boolean variables, there are  $2^{2^n}$  different Boolean functions.

If Boolean functions are 4, then

$$2^{2^n} = 4$$

$$2^{2^n} = 2^2$$

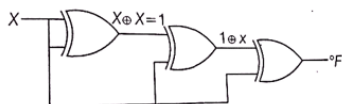
On comparing both sides, when base are equal

$$2^n = 2$$

$$2^n = 2^1$$

So,  $n = 1$

6. (d)



The complement of ( $F = 0$ ) is ( $F = 1$ )

7. (d)

For the case of  $n$  Boolean variables, there are  $2^{2^n}$

Boolean expression can be formed.

If value of Boolean variable is 3.

Then Boolean expression will be  $2^{2^n}$

$$= 2^8 - 256$$

8. (a)

Binary representation of 47 = 101111

8 bit representation of 47 = 00101111

Binary representation of 38 = 100110

8 bit representation of 38 = 00100110

$(47)_{10} - (38)_{10} = 00101111$

$-00100110$

1's complement = 11110110

2's complement = 11110110

$$\begin{array}{r} + \quad 1 \\ 11110110 \\ \hline \end{array}$$

9. (a)

The memory unit which directly communicates with the CPU is known as primary memory. Primary memory or main memory refers to physical memory that is internal to the computer. Every program you execute and every file you access must be copied from a storage device into main memory.

10. (c)

Dynamic RAM consumes less power and slower than static RAM, because of the needed of charging and discharging of capacitor in DRAM. It consumes less power than SRAM also slow too.

11. (b)

The time required for the fetching and execution of one simple machine instruction is CPU cycle.

The speed of a computer processor or CPU, is determined by the clock cycle, which is the amount of time between two pulses of an oscillator. The higher number of pulses per second, the faster the computer processor will be able to process information.

12. (b)

$$x + x'y$$

$$= x.1 + x'y$$

$$= x(1 + y) + x'y$$

$$= x + xy + x'y$$

$$= x + y(x + x')$$

$$= x + y.1$$

$$= x + y$$

$$[\because A.1 = A]$$

$$[\because A' + A = 1]$$

$$[\because A + A' = 1]$$

13. (c)

Binary equivalent of

$$(4ACO)_{16} = 0100101011000000$$

Binary equivalent of

$$(B53F)_{16} = 1011010100111111$$

Now, perform the XOR operation on  $(4ACO)_{16}$  and  $(B53F)_{16}$

$$(1111111111111111)_2$$

It is equivalent of  $(FFFF)_{16}$ .

14. (b)

Pipe-lining in the process of improving the performance of the system by processing different instructions at the same time, with only one instruction performing one specific operation.

15. (d)

Accumulators are used by ALU (Arithmetic Logic Unit) to store the intermediate results. The ALU is the computational center of the CPU. It performs all mathematical and logical operations.

16. (d)

$$1 \text{ Giga Byte (GB)} = 1024 = 10^{10} \text{ MB}$$

$$1 \text{ Tera Byte (TB)} = 2^{10} \text{ GB}$$

$$1 \text{ Peta Byte (PB)} = 2^{10} \text{ PB}$$

$$1 \text{ ExaByte (EB)} = 2^{10} \text{ PB}$$

$$= 2^{10} \times 2^{10} \text{ TB}$$

$$= 2^{10} \times 2^{10} \times 2^{10} \text{ GB} = 2^{30} \text{ GB}$$





# MAARULA CLASSES

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17. (b)

The cache memory is more effective because of locality of reference. This means that the cache depends on the locations in the memory that is referenced often.

18. (a)

Registers are fastest means of access for CPU.

Registers are the small memory locations which are present closet to the CPU.

19. (a)

To fetch data from the secondary memory, MAR (Memory Address Register) is used. MAR is a CPU register that either stores the memory address from which data will be fetched to the CPU or the address to which data will be sent and stored.

20. (d)

Number of steps =  $2n^2$

Speed of computer =  $10^6$  per second

For  $n = 10000(10^4)$

$$\therefore \text{Time} = \frac{\text{Number of Steps}}{\text{Speed of Computer}}$$

$$= \frac{2n^2}{10^6} = \frac{2 \times (10^4)^2}{10^6} = \frac{2 \times 10^8}{10^6}$$

$$= 2 \times 10^2 = 200 \text{ sec.}$$

